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Tom Keeline, Tyler Kirby

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Latin Vocabulary and Reading Latin: Challenges and Opportunities*

TOM KEELINE

Washington University in St. Louis

TYLER KIRBY

Independent Scholar

ABSTRACT: Scholars agree that you need to know 95–98% of the words in a text in order to understand it. Using a digital and statistical analysis of a four-million-word corpus, we quantify the challenge of reaching this threshold in classical Latin. The vocabulary distribution has a long tail: to read with fluency you'd have to learn a lot of uncommon words. Because of the nature of the extant classical Latin corpus, this is an almost impossible task today. But Latin learners shouldn't despair. Instead, we should adjust our expectations, our teaching, and our reading practices. We make suggestions for possible changes.

KEYWORDS: Latin pedagogy, vocabulary, second language acquisition, digital and statistical analysis, reading Latin

INTRODUCTION

A FEW YEARS AGO MARY BEARD admitted in a blog post that she can't just pick up a book written in Latin and read it.¹ The internet exploded. "Fraud!" the headline might have screamed. "Celebrity Cambridge Classicist confesses she can't read Latin." Disgust and incredulity filled the ether: "And she calls

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¹ Beard 2016; cf. Keeline 2019a, which generated less online furor.

herself a classical scholar?”² Partly this was an unreasonable reaction to Mary Beard herself, but partly it grew out of a not unreasonable sentiment that a professor of Classics really should be able to read Latin, just as a professor of French can read French. And so internet pundits around the world formed up ranks and took a stand: maybe Mary Beard can’t read Latin, but surely it’s possible to do so. Latin, they maintained, is a language just like any other.

For one of the authors of this paper—a so-called professional Latinist who has studied the language for more than two decades—Mary Beard’s experience sounds all too familiar. Despite his best efforts, he still can’t read classical Latin as well as he “should” be able to; over the past twenty years it’s gotten easier, but it still hasn’t gotten easy. While he hopes he can work his way through just about any Latin text, he can’t read an author like Tacitus the way he reads Edward Gibbon. Not even close.

Perhaps the internet idealists are right: even if many of us supposed professionals can’t read classical Latin the way we read our native language, maybe with the right training *someone* could. The Romans did. Some people may indeed be doing so today. Perhaps those of us who still struggle with Latin despite decades of study should be wary about generalizing from our experience; as Sallust put it, *quae sibi quisque facilia factu putat, aequo animo accipit, supra ea veluti ficta pro falsis ducit* (“Everyone is happy to believe things that they think they can easily do themselves, whereas things beyond their own abilities they consider to be made-up falsehoods,” *Cat.* 3.2). And yet, while we welcome correction from fluent first-time readers of Tacitus, we suspect that a lifetime of struggle is the norm. And so we want to state plainly and publicly what many students and scholars of Latin feel but seldom seem to acknowledge—namely, that reading classical Latin is just hard.³ More importantly, we want to explain *why* it’s hard, even for many of us who have been doing it for a long time. Latin is indeed a language, but it’s not “just like any other language.”

² For a typical response along these lines, albeit expressed with more thought and nuance, see Foreman 2019. A Google search will reveal much more vitriolic reactions.

³ This might be our profession’s “dirty little secret” (cf. Kitchell 2000), but it shouldn’t be. Some Classicists do acknowledge the difficulty of reading Latin, only to promise that it can be overcome by this or that system (e.g., Hoyos 1997: 11: “Anyone can read Latin directly from the page ... all that is needed is a knowledge of accident and syntax, along with common sense and ... a commitment to practicing. The methods are easy to understand and straightforward to apply”). Such rhetoric is so common in online Latin communities that it’s almost not worth citing examples (but cf., e.g., the blog post cited in n2 above), and it’s also the general rhetoric of university course descriptions (cf. n35 below).

Modern readers face any number of obstacles when trying to read classical Latin: complex morphology and syntax, “strange” word order, unfamiliar subject matter, foreign cultural and historical context, unbridgeable gaps of space and time, and so on *ad infinitum*. But the most fundamental barrier to understanding Latin has long been overlooked, or at least underestimated: the Latin vocabulary. Contemporary scholars of reading and second language acquisition (SLA) agree that you need to know 95–98% of the words in a text in order to read it with understanding.⁴ For students reading Latin texts today, this is all but impossible; even for professional Classicists, it constitutes a very high bar.

In the first part of our paper, we quantify how high the 95–98% bar is for classical Latin. Using a digital analysis of a four-million-word corpus of Latin written 100 B.C.E.–200 C.E., we show what words you’d need to know to have a chance of reading a given author fluently, i.e., the baseline level of vocabulary knowledge to even try reading an author, leaving aside grammar and syntax and cultural context and so forth. The good news is that there is a common core of words that occur frequently, and this common core is very helpful to learn and isn’t huge. The bad news is that the distribution of Latin words has an extremely long tail, and to read with fluency you would have to learn a lot of words that do not occur very frequently at all. Unfortunately, because of the size and lexical complexity of the extant corpus, this is an almost impossible task for readers of classical Latin. It is the nature of the corpus that sets classical Latin apart; this is what makes Latin different from French or any other modern language. And so in practice we simply will not be able to “read” most classical Latin the way that you might read a novel in English or Japanese.⁵

⁴ See, e.g., Laufer 1989, 1992, 2021; Hu and Nation 2000; Nation 2006; Schmitt 2008; Schmitt, Jiang, and Grabe 2011. The definitions of “know a word” and “understand a passage” vary—both are always on a continuum—and pose ultimately intractable philosophical problems (e.g., on the meaning of “meaning”: see Speaks 2019 for an introduction). SLA researchers assess word knowledge in students by self-report (e.g., “mark the words you know / don’t know”) and/or by various other vocabulary tests; they assess understanding of passages by reading comprehension tests, with an arbitrary cutoff for “understanding” (e.g., a test score of 55%). For more detailed attempts to qualify what it means to know a word (denotation, connotation, collocations, and much more), see esp. Nation 2013: 44–91.

⁵ In this article we are concerned only with this kind of reading, the way you read literature with ease and pleasure in a modern language that you know well. Other approaches—reading for the gist of a passage, say, or extracting meaning from a text by careful philological study—are of course possible, but we are interested in why the Spanish professor can pick up an Argentinian book and read it with excellent comprehension, whereas the Latin professor probably cannot do the same for a previously unseen book written in first-century Rome.

But in the second part of our paper, we argue that students and scholars of Latin shouldn't just despair. Instead, we should adjust our expectations, our teaching, and our reading practices. We make some suggestions for possible changes. It may be impossible to read Latin with true fluency, but it's certainly possible to get better at it.⁶

PART I: THE 95-98% CHALLENGE IN LATIN

95-98%: "Really?"

It might come as a surprise to be told that you need to know 95-98% of the words in a text in order to read it with understanding. It sounds counterintuitive. "Forget the SLA studies," someone might say. "Surely if I know 80% or 90% of the words in a text I can understand it."⁷ Can you?

Consider the case where you are probably a native (or near-native) speaker and know the grammar and cultural context perfectly: consider the case of a contemporary English short story. The following are three consecutive extracts from "The Ones Who Walk Away from Omelas," a delightful piece of speculative fiction by Ursula K. Le Guin.⁸ In the first gobbet we've replaced six of the

⁶ *Mutatis mutandis*, everything we say here can probably be said for ancient Greek too, except Greek is likely to be even harder, because "ancient Greek" embraces much more chronological and dialectal variety than the fairly standardized "classical Latin" (although Greek's common lexical core may actually account for a larger percentage of its words: see Major 2008).

⁷ Cf., e.g., Rydberg-Cox and Mahoney 2002: 148 on the Perseus Project Word Study Tool: "For an intermediate-level class, beginning to read unadapted texts, you can generate a word list. ... Select five or six texts ... sort by weighted frequency, and request the top 50% or 60%. The list will be quite short, probably 200-500 words ... and most of them will be familiar already ... *Once your students know all these words, they can begin reading, confident that they will know half to two-thirds of the words on a typical page. Many students are pleased and reassured to find out that the core vocabulary of ordinary Latin or Greek prose is so small*" [our emphasis]. So too, e.g., Distler 1962: 117, suggesting 80-90% ought to be sufficient; indeed, until SLA researchers began studying these questions in the early 1980s, such optimism was widespread among language teachers more broadly (see Laufer 2021: 238).

⁸ First published in 1973, "The Ones Who Walk Away from Omelas" won the Hugo Award in 1974 and was republished in 1975 in the collection *The Wind's Twelve Quarters*; it has been frequently anthologized since.

original 111 words with non-words that nevertheless mirror the original (e.g., verbs are still recognizably verbs).⁹ You thus know about 95% of the words:¹⁰

With a crenamara of bells that set the swallows soaring, the Festival of Summer came to the city Omelas, bright-towered by the sea. The gindre of the boats in harbor silletted with flags. In the streets between houses with red roofs and painted walls, between old moss-grown gardens and under brills of trees, past great parks and public buildings, processions moved. Some were decorous: old people in long stiff robes of mauve and grey, grave master workmen, quiet, lullious women carrying their babies and chatting as they walked. In other streets the music beat faster, a shimmering of gong and temello, and the people went dancing, the procession was a dance.

You probably do have a sense for what is going on here. In some cases you might be able to guess with a fair degree of certainty what an unknown word means: “crenamara,” for example, is probably something like “tolling” or “pealing.” (Le Guin in fact wrote “clamor.”) Deducing the meaning of unknown words from context is an important skill, and it’s a completely natural one: as we read, we’re constantly drawing inferences based on what we already understand.¹¹ But already at the 95% level it’s becoming very hard to make such deductions. Even if the general run of the passage seems clear, a word like “gindre” is likely to remain opaque. And if you’re like us, you’ll have the feeling that if you only know 95% of the words in a passage, while you might be able to get meaning out of it, you probably wouldn’t want to read paragraph after paragraph and page after page of such passages for fun. There is already too much missing; the picture is already blurring at the edges.

Try 90%:

Children pixed in and out, their high calls rising like the swallows’ dossing rotels, over the music and the lagathy. All the processions wound towards the north side of the city, where on the great malarkin called the Green Fields

⁹This is a common technique in assessing readability with unknown vocabulary items: see, e.g., Hu and Nation 2000; Milton 2009: 47–50; Schmitt, Jiang, and Grabe 2011: 31.

¹⁰Rather than three consecutive short passages with a decreasing number of known words in each, one might alternatively present one longer passage in three versions, each with an increasing number of known words. Such a procedure has the disadvantage of making the 95% version look artificially better (it is a lot better by comparison to 80%!), and it can also introduce an element of retrospective false confidence (“Oh, I see what that word must have meant”).

¹¹This is one of the reasons that it’s so hard to read unfamiliar material: you are no longer able to make successful deductions and inferences, and so misunderstandings based on false or unfulfilled expectations tend to compound.

boys and girls, foresome in the bright air, with keeny feet and ankles and long, lithe arms, exercised their restive horses before the race. The horses wore no nabbin at all but a wyself without bit. Their manes were braided with grimfels of silver, gold, and green. They flared their nostrils and jiffited and boasted to one another; they were vastly excited, the horse being the only velit who has adopted our ceremonies as his own.

You know 90%—nine out of every ten—words in this passage. Can you get meaning out of it? Even if you have the gift of second sight, you'll probably concede that you're missing something in almost every sentence, and that the passage as a whole feels murky.

At 80% “murk” becomes impenetrable darkness:

Far off to the north and west the uretils stood up half ritiling Omelas on her bay. The air of zylon was so ilthet that the snow still milpitting the Eighteen Peaks burned with white-gold miscrot across the jem of belet air, under the dark blue of the sky. There was just enough welth to make the cellers that grilped the racecourse cralt and menapper now and then. In the silence of the broad ulpet meadows one could hear the howet winding through the city welfts, farther and nearer and ever tiroling, a cheerful relipt critellian of the air that from altel to altel trembled and gefted together and broke out into the great joyous farfelling of the manets.

Such a passage lies beyond the abilities of even the most skillful diviner.

Now consider the case of Latin. You are no longer a native speaker with complete mastery of grammar and all the other “known” words in a text. You have, at best, a very imperfect understanding of the cultural context. And you face countless other difficulties besides. In brief: your task is much, much harder. That 95% bar represents an absolute minimum for reading comprehension; 98–99% is probably more realistic.¹² Grammar and cultural context and so forth are all important for reading with understanding, but vocabulary is essential: without words, you can't read.

Quantifying the challenge in Latin: The data

We wanted to quantify how hard it is to know 95–98% of the words in a Latin passage. We thus developed a set of algorithms and modules to answer this question and formalize mathematically our intuition that vocabulary is a

¹²SLA researchers seem to be converging on the idea that 95% represents a minimum for “study reading,” while 98–99% coverage is needed for pleasure reading; see, e.g., Laufer 2021: 238–39 and the other references in n4.

major reason why classical Latin is hard to read.¹³ First, we created a corpus of all the “classical” Latin authors from 100 B.C.E.–200 C.E. who happen to be available on The Latin Library website, about four million words of Latin.¹⁴ We excluded older texts because archaic forms foxed our digital parsing tools; we excluded more recent texts simply because we could not freely acquire a comprehensive or representative sample of later Latin. These limitations are thus arbitrary and in some ways unfortunate—they certainly don’t reflect value judgments—but they do have the virtue of representing a relatively stable and standardized form of literary Latin, and they correspond to the canonical texts most commonly taught in Latin classrooms as well.

The corpus was downloaded and preprocessed extensively, imposing standard orthography and formatting and segregating certain problematic features.¹⁵ This done, we used a neural network-based lemmatizer to reduce the texts into a series of lemmata, i.e., dictionary headwords;¹⁶ lemmata, rather than individual words or collective word families, seem like the most

¹³ All of our code and documentation are open source and available online (Kirby 2022); readers who would like to conduct their own queries or reproduce our results can also use our website app, the Latin Vocabulary Data Explorer: <https://doi.org/10.7936/s8wg-vx31>.

¹⁴ We used these texts because they are freely available with a minimum of formatting; the inevitable occasional textual errors are insignificant in a corpus this large. Full details on our corpus can be found in our code and documentation, but note that fragmentary works are usually excluded and Pliny the Elder includes only *Nat.* 1–4.

¹⁵ For full details, see our code and documentation. Note that in most vocabulary research, proper names are omitted on the grounds that readers recognize them, but successful named entity recognition proved a challenge to program for Latin (the existing modules were not successful in our tests, nor were our preliminary attempts using machine learning; cf. Erdmann et al. 2016). We thus simply removed from our corpus all capitalized words which do *not* begin a sentence. This dampens their effect on our corpus significantly. We would argue that those that are left—i.e., those that happen to start a sentence—represent a reasonable compromise, because if you don’t know what *Gallia* is (say), you can’t understand the first sentence of Caesar’s *Gallic Wars*. Many Latin proper names won’t be automatically recognized by readers and hence really do constitute vocabulary items.

¹⁶ Lamontpy (Lee 2020), which in our tests proved extremely accurate. No automated lemmatizer can be perfect, but again, in a corpus of four million words, occasional mistakes do not affect overall results. Lamontpy will always return a dictionary headword from Lewis and Short, i.e., it will never lemmatize to a nonsense form. For recent discussion of Latin lemmatizers (although not of Lamontpy), see Burns 2020 and Celano 2020. Another promising approach to lemmatization is Pie (LASLA 2021), using data from the Laboratoire d’Analyse Statistique des Langues Anciennes. For comparison of Greek lemmatizers, see Vatri and McGillivray 2020.

sensible units to analyze.¹⁷ Finally, the data were tabulated and analyzed and presented in a variety of formats.

The headline findings are straightforward and not, in and of themselves, surprising.

1. Yes, there is a common core of Latin vocabulary. Once you know 1,308 lemmata, for example, then you know 80% of the words in our corpus.¹⁸ Unfortunately, 80% isn't enough to read anything (cf. "there was just enough welth to make the cellers that grilped the racecourse cralt and menapper now and then"). Perhaps even more unfortunately, students of Latin tend to learn far fewer words. Wheelock, for example, has a vocabulary of only 843 words. Indeed, of commonly used textbooks, only Hans Ørberg's *Lingua Latina per se illustrata* meets even this threshold (1,811 words)—and no student is likely to have mastered all these words while using the book in a typical introductory language sequence.¹⁹
2. The distribution has an extremely long tail:

TABLE 1. OVERALL CORPUS

<i>Number of lemmata known</i>	<i>Percent of words in corpus known</i>
1,308	80%
2,668	90%
4,319	95%
6,670	98%

¹⁷ Some SLA research does prefer "word families," which would group *celeriter* (#1274 on our frequency list) with *celer* (#1439), but problems immediately arise: what about *accelero* (#5432)? Do *aptus* (#763) and *ineptus* (#3197) belong together? Even apparently straightforward verbal compounds vary widely: if you know *venio* ("come"), you can probably deduce *advenio* ("come to" → "arrive"), but *convenio* ("fit"), *evenio* ("happen"), and *invenio* ("find") will remain obscure (although their etymology may help you remember such words once you've learned them: cf. n37 below). Unlike English, the highly inflected nature of Latin makes almost any lemma already a large word family (e.g., *fero* includes *fertur* and *tulisset* and *laturis*, more than 200 discrete word forms). It thus seemed not only programmatically easiest but methodologically best to work with lemmata.

¹⁸ These results are broadly in alignment with other attempts at quantifying the Latin vocabulary, most influentially those of Diederich 1939, but the further down you go on the word frequency table, the more the specific composition of the corpus matters. For earlier surveys of the Latin vocabulary, see Delatte et al. 1981; Muccigrosso 2004: 413–19; Francese 2021; Gruber-Miller and Mulligan 2022; on Greek vocabulary, see Major 2008.

¹⁹ If Ørberg's vocabulary represented the 1,811 most frequent lemmata in our corpus—which it does not—then a student who had mastered it perfectly would know about 84.8% of the words in our corpus. For further details on the vocabulary of Latin textbooks published prior to 2004, see Muccigrosso 2004.

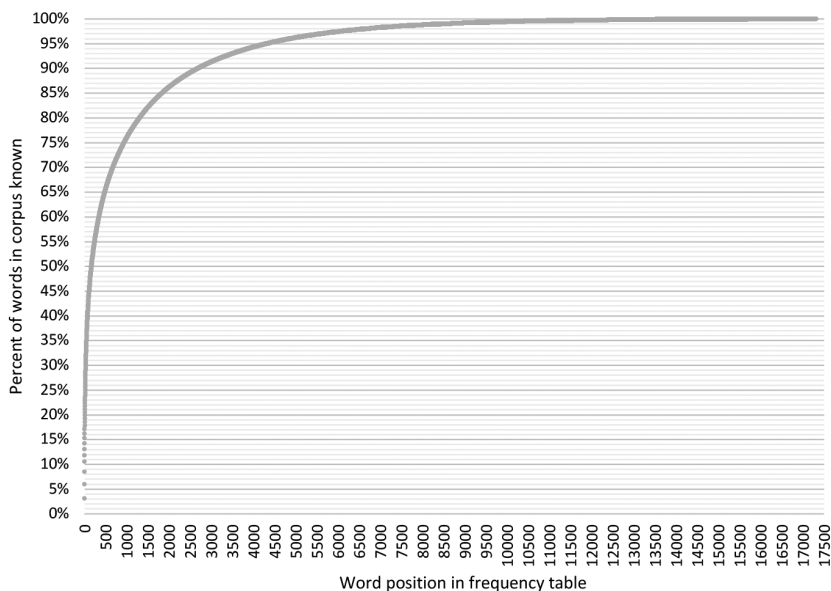


Figure 1. Vocabulary Distribution in our Corpus

To get to 90%, you'll need to know about 2,688 words—i.e., an additional 1,380 words from the 80% level. To get to 95%, you'll need some 4,319 words. And to reach that magical 98% mark for our corpus, you need a perfectly internalized working vocabulary of 6,670 words. And these are words that do not show up very often: at the 6,670-word mark, we are talking about lemmata that occur 30 total times in our corpus of about four million words.

While the specific numbers here are new, the results in general are expected. The distribution of vocabulary items in a language tends to conform to Zipf's Law, which states that the frequency of any word is inversely proportional to its rank in the frequency table (for a graphical illustration, see figure 1 above).²⁰ Put simply, in any natural language, a few words tend to be really common, and everything else is relatively rare. In English, for example, 135 lemmata will account for about half the words in a corpus of over a million words.²¹ In our Latin sample, once you know 169 lemmata, you know 50% of the words in the corpus.

But this distribution poses unique challenges for anyone who wants to learn to read Latin. Pick any author and the difficulties immediately become

²⁰ On Zipf's Law, see, e.g., Piantadosi 2014.

²¹ The Brown Corpus: see Francis and Kučera 1967.

apparent. Take Catullus, for example. In our adjusted Catullan corpus of 12,118 words, we have 2,770 lemmata. This might not seem like an excessive number. If you know 874 lemmata, you're at 80% of the corpus. Knowing 1,559 lemmata gets you to 90%; 2,165, 95%; 2,528, 98%:

TABLE 2. CATULLUS: 12,118 WORDS

<i>Lemmata known</i>	<i>Percent of words in corpus known</i>
874	80%
1,559	90%
2,165	95%
2,528	98%
2,770	100%

While 2,500 words is far beyond the range of most students—certainly most intermediate students—you could hope that it would be well within the range of their teachers. But there is a further problem: Catullus's 2,770 words are not the 2,770 most common words in the Latin language. You can't read Catullus unless you know 98% of *his* words. If you knew the 3,000 most common words in our overall corpus of classical Latin, you would still only know 86.16% of the words in Catullus; his poetry would look like little more than word salad in stichic form.

The first column in the following table (table 3) shows the percentage of words you would know in the individual authors of our corpus if you had a vocabulary of the top 3,000 words in the corpus as a whole. The authors are ranked from most to least lexically challenging.²² (The second and third columns are the results of a Monte Carlo-style reading simulation, which we will describe below.)

In fact, if you learned even the top 5,000 words in our Latin corpus, you would still only know 91.79% of the words in Catullus. Suddenly the magnitude of the challenge starts to become clear: if you know the 5,000 most common words in our corpus, you still don't quite know 98% of even Caesar, Sallust, or Cornelius Nepos.²³

²² Treating authors like (e.g.) Cicero, Horace, and Seneca the Younger as single entities has its advantages and disadvantages, but for our present purposes we are only interested in the picture's overall shape, not its details. Our code can easily be adjusted to subdivide authors into their individual works.

²³ You are admittedly very close. Note too that a comparison of Latin prose authors shows how vocabulary, while essential for reading, is not the only thing affecting how

TABLE 3. VOCABULARY: 3,000 MOST COMMON WORDS KNOWN

	<i>Percentage of known words in author</i>	<i>Monte Carlo– percentage of known words in author</i>	<i>Monte Carlo standard deviation in author</i>
Persius	79.79%	79.94%	2.96%
Columella	83.27%	83.52%	4.95%
Apuleius	83.30%	79.53%	4.06%
Calpurnius Siculus	84.05%	82.42%	4.06%
<i>Carmina Priapea</i>	84.65%	84.19%	2.59%
Pliny the Elder	85.83%	88.47%	3.91%
Martial	85.89%	86.79%	5.08%
Horace	85.92%	86.66%	2.83%
Vitruvius	85.94%	85.20%	4.78%
Catullus	86.16%	87.56%	4.53%
Petronius	86.38%	84.55%	4.37%
Suetonius	86.39%	86.81%	2.64%
Statius	86.58%	87.25%	2.33%
Juvenal	86.72%	85.78%	3.93%
Silius Italicus	87.21%	87.92%	3.24%
Germanicus	87.40%	86.41%	2.52%
Grattius	87.53%	89.00%	3.27%
Vergil	88.19%	85.46%	4.24%
Varro	88.90%	89.75%	4.18%
Valerius Flaccus	88.92%	91.06%	2.73%
Tacitus	89.07%	88.50%	2.55%
Frontinus	89.50%	89.87%	2.30%
Propertius	89.56%	92.39%	2.38%
<i>Res gestae</i>	89.71%	92.35%	2.10%
Valerius Maximus	89.84%	88.67%	2.84%
Lucan	90.09%	87.66%	3.58%
Tibullus	90.63%	90.73%	3.46%
Ovid	90.98%	93.26%	1.99%
<i>Ilias Latina</i>	91.03%	91.63%	1.71%
Manilius	91.13%	90.19%	3.52%
Hyginus	91.34%	89.79%	3.19%
Velleius Paterculus	91.41%	90.21%	2.39%
Pomponius Mela	91.48%	91.11%	2.59%
Pliny the Younger	91.75%	90.31%	2.82%
Curtius Rufus	91.86%	92.24%	2.09%
Seneca the Younger	92.28%	92.84%	2.55%

Table 3, continued

	<i>Percentage of known words in author</i>	<i>Monte Carlo– percentage of known words in author</i>	<i>Monte Carlo standard deviation in author</i>
Quintilian, <i>Decl.</i>	92.61%	93.02%	1.53%
Asconius	92.95%	93.90%	3.13%
Seneca the Elder	92.96%	93.05%	2.21%
Quintilian, <i>Inst.</i>	92.98%	93.28%	1.98%
Livy	93.26%	93.53%	1.78%
<i>Corpus Caesarianum</i>	93.82%	94.00%	2.34%
Sallust	94.08%	93.77%	1.19%
Cicero	94.18%	96.26%	1.67%
Sulpicia	94.26%		
Caesar	94.36%	93.49%	3.53%
Cornelius Nepos	94.84%	95.13%	1.27%

TABLE 4. VOCABULARY: 5,000 MOST COMMON WORDS KNOWN

	<i>Percentage of known words in author</i>	<i>Monte Carlo– percentage of known words in author</i>	<i>Monte Carlo standard deviation in author</i>
Persius	87.95%	88.80%	3.34%
Apuleius	91.12%	89.95%	3.26%
Columella	91.42%	91.47%	3.83%
<i>Carmina Priapea</i>	91.77%	90.79%	2.09%
Catullus	91.79%	93.23%	2.94%
Calpurnius Siculus	92.08%	92.03%	2.42%
Pliny the Elder	92.18%	94.69%	2.40%
Martial	92.48%	92.80%	4.11%
Vitruvius	92.52%	92.35%	3.09%
Horace	93.27%	94.18%	1.89%
Juvenal	93.28%	92.51%	2.89%
Petronius	93.34%	92.72%	3.67%
Suetonius	93.38%	93.52%	1.92%
Varro	94.06%	94.39%	3.02%
Grattius	94.07%	94.80%	2.25%
Statius	94.30%	95.52%	1.49%
Germanicus	94.41%	92.96%	1.81%

Table 4, continued

	Percentage of known words in author	Monte Carlo– percentage of known words in author	Monte Carlo standard deviation in author
Vergil	94.53%	92.59%	3.48%
Silius Italicus	94.72%	95.38%	1.96%
<i>Res gestae</i>	95.19%	96.94%	1.59%
Valerius Flaccus	95.33%	96.58%	1.80%
Propertius	95.52%	97.21%	1.05%
Frontinus	95.63%	95.51%	1.72%
Tacitus	95.64%	95.45%	1.22%
Hyginus	95.86%	95.14%	1.89%
Valerius Maximus	96.12%	95.42%	1.30%
Ovid	96.14%	97.09%	1.47%
Lucan	96.22%	95.12%	3.90%
Pliny the Younger	96.54%	96.01%	1.26%
Velleius Paterculus	96.55%	96.45%	1.61%
Manilius	96.70%	96.44%	1.50%
Tibullus	96.76%	96.80%	1.49%
<i>Ilias Latina</i>	96.86%	97.52%	1.33%
Pomponius Mela	96.90%	96.76%	1.16%
Seneca the Younger	96.95%	97.03%	1.60%
Quintilian, <i>Inst.</i>	97.03%	96.84%	1.33%
Asconius	97.13%	97.67%	1.21%
Curtius Rufus	97.36%	97.83%	1.13%
Seneca the Elder	97.39%	97.43%	1.30%
Quintilian, <i>Decl.</i>	97.47%	97.30%	0.99%
Livy	97.49%	97.77%	1.71%
Cicero	97.49%	98.74%	0.94%
<i>Corpus Caesarianum</i>	97.63%	98.08%	1.70%
Caesar	97.86%	97.35%	2.63%
Sallust	97.95%	98.23%	0.89%
Cornelius Nepos	97.96%	98.08%	1.27%
Sulpicia	98.36%		

Another way to approach this question is to ask: if you open up an author to a random page, could you read it? I.e., what percentage of the words on the page would you know? On any individual page, of course, your results may vary, and it's precisely that variance that is of interest. The average percentage known on individual pages will, over a large number of trials, converge on the percentage known in the entire author, because if you open every single page of the author, then you're simply calculating the result for the author as a whole. But the variance may have something to tell us: do we see wild swings where you might know 80% of the words on one page and 99% on the next?

We tested this question in a Monte Carlo-style experiment, taking fifteen random but continuous 250-word samples from each author and calculating the mean percentage of words known and its standard deviation (the second and third columns of tables 3 and 4 above).²⁴ In brief: the variance is not large, but especially at lower levels of vocabulary knowledge, it is large enough that you might notice an effect. At the 3,000-word level, in our Monte Carlo experiment you know 93.49% of the words on an average page of Caesar. But the standard deviation is 3.53%. That means, in effect, that in our sample of fifteen pages, it's likely that there was at least one page where you knew 97% of the words. And it's also likely that on at least one page you knew less than 90% of the words. This is a statistical accounting for an experience we've all had: sometimes we're happily reading along and understanding everything perfectly, then we turn the page and almost immediately become lost. Even at higher levels of vocabulary knowledge, like 5,000 words, you're still likely to come across occasional pages even in Caesar where you know less than 95% of the words.

Admittedly, no one needed a statistical analysis to determine that Persius is almost impossible to read, although our numbers do quantify just how impossible he is. Nor will Latinists be surprised to hear that Horace and Juvenal are hard, or Petronius and Apuleius. But the unfortunate reality is, on this analysis, just about every extant Latin author will prove impossible for students to read fluently, and most will offer some challenge even to full-time Classicists.

difficult an author is: Cicero, for example, is all but indistinguishable from Cornelius Nepos in terms of vocabulary even at the 3,000-word level.

²⁴ In a Monte Carlo simulation, you perform repeated experiments while changing a key variable, then observe the distribution of results. (Sulpicia is excluded from our calculations here because her corpus is too small for repeated sampling.)

Why is Latin uniquely hard?

But why is Latin different from English or any other language? In a sense, it's not. Scholars of English reading generally estimate that you need to know 8,000–9,000 vocabulary families to ensure 98% coverage of authentic English texts (excluding proper nouns).²⁵ This is arguably harder than Latin: in Latin, at 6,500 lemmata, you're over 98% for all commonly read Latin prose; at 7,500 lemmata, you're over 99%; by 8,500 lemmata, you're at the 98–99% threshold for all but the most recalcitrant and *recherché* authors. In this sense it should be easier to master Latin vocabulary than English vocabulary; you can read more with fewer words. The real difference between English and Latin, at least for the purpose of reading English and Latin literature, is the available sources of input. While English comes in a variety of levels, all extant classical Latin is just different gradations of “hard.”

Much SLA scholarship—and common sense—suggests that the best way to learn a new word is to be exposed to it repeatedly in meaningful contexts that you can understand.²⁶ This is all but mathematically impossible with classical Latin—which is the only source of input found in most intermediate and advanced Latin classrooms. The reason is twofold. The first problem follows from the word “repeatedly”: in our four-million-word corpus, lemmata from position 3,827 onwards in the frequency table—which happens to be the 93.88% mark—occur 100 times or less, i.e., their relative frequency is .000025 or less. Words at the 95% level occur 81 times or less in our corpus; words at the 98% level occur 30 times or less. Estimates of the number of meaningful occurrences of a word required to internalize it vary, ranging from six to more than twenty, but let's pick twelve: assuming a random distribution of vocabulary (a false but simplifying assumption), you would need to read through about *half* of our total corpus in order to internalize all of these words by reading.²⁷ But the second problem follows hard on the heels of this one: you can't actu-

²⁵ See, e.g., Nation 2006; Laufer 2021: 238 with further references. On “families” vs. lemmata, see n17 above.

²⁶ See, e.g., Barcroft 2016: 24–28.

²⁷ For details on repetition and vocabulary learning, see Nation 2013: 451–58, synthesizing previous studies; further Schmitt 2008: 343; Nation 2014: 2–3. The number twelve is not completely arbitrary: it's the suggestion of Nation 2014 as a “moderately safe goal” and “the minimum,” allowing “the opportunity for several dictionary look-ups, several unassisted retrievals, and an opportunity to meet [a] word in a variety of contexts” (3). Nation 2014: 7–8 does indeed recommend that an advanced language learner seeking to master English vocabulary read three million words (or about twenty-five novels)—in a single year, and this after already having read eight million words in previous years!

ally read half of our corpus, or at any rate you can't read half of our corpus "in meaningful contexts that you can understand," because you're trying to learn thousands of relatively rare words all at once, and you can't learn one without knowing the others. You are thus trapped in a web of dependencies. Philologists spend much of their lives hacking through this web, looking up word after word in the dictionary and garnering a certain level of vocabulary knowledge by brute force, but this is, at best, slow and painful.

What these difficulties really point up is a much broader problem related to Latin input: we lack level-appropriate texts. In order to develop an accurate mental representation of a language, learners need massive amounts of comprehensible input.²⁸ This is the reason for communicative classrooms in modern languages. This is the main reason that immersion in a language in a foreign country is so effective.²⁹ This is why our modern language colleagues cultivate extensive reading and have their students watch movies and listen to music and on and on and on: the goal is always to increase input. No one would give EFL students—students learning English as a Foreign Language—Shakespeare in their third semester. You give such students texts that they can understand and enjoy, i.e., you give them compelling and comprehensible input. You can find virtually unlimited English spoken and written by native speakers, on any subject, at any level of difficulty, including the beginning and intermediate levels. But when it comes to classical Latin, we may have literally *nothing* that fits into those categories.³⁰

Modern language instructors adjust the difficulty of their texts to suit their students' abilities. Teachers of Classics, by contrast, committed to reading "real Latin" as soon as possible, tend to reason something like this: "Well, it takes my students two hours to 'read' two paragraphs, and so I'll only assign them

²⁸ "Comprehensible input" is most associated with the theories of Stephen Krashen (e.g., Krashen 1982), but you don't need to subscribe to every detail of his hypotheses to understand that input is the *sine qua non* of language acquisition. You can't learn anything without input.

²⁹ Although much of an immersive environment is incomprehensible, part of it is understood, and as learners build on that foundation they come to understand ever more of what they hear. Necessity of course plays a strong role in motivating learners in such situations.

³⁰ While in this paper we discuss the lack of "comprehensible" input in Latin, the lack of "compelling" input should not be underestimated. The subject matter, genre, and cultural and aesthetic sensibilities of almost all classical Latin literature are fundamentally different from our modern expectations, and some contemporary readers will have little interest in turning pages and "finding out what happens next" in almost any classical Latin author. This subject, however, deserves a separate discussion.

two paragraphs.” But from an SLA perspective, this is madness. If you need massive amounts of comprehensible input, on such a system you will *never* get it, or at least not in any reasonable amount of time. And so you will never be able to internalize vocabulary successfully. Consider a class that reads all of Catullus over a semester: out of 2,770 Catullan lemmata, 2,135 occur three times or fewer, of which 1,370 occur only once in Catullus. There simply isn’t enough exposure to these words to ensure that students can actually learn them. (And remember, you can’t read Catullus fluently without learning something like 2,500 of these 2,770 lemmata.) Perhaps students in such a class can translate the requisite passages into English—through some combination of memorization and guessing and actual knowledge—but they’re unlikely to be able to retain most of these relatively infrequent vocabulary items and transfer them to reading in future semesters. And this inability has nothing to do with students’ virtues or lack thereof. What we’re asking them is, at least for most people, simply impossible.

Students of Classics, in turn, develop coping mechanisms. Consider the famous and widely deprecated “find the verb.”³¹ Well, if you’re confronted with an elegiac couplet and need to look up half of its words, it just makes sense to start with finding the verb. Because looking up the verb is going to give you information about what kind of objects and other words you should expect in the sentence; it’s the single word most likely to help you figure out what the rest of the sentence means. “It’s poetry,” you might say to yourself: you’re not getting a lot of meaning from the word order anyway. Such a procedure is an entirely rational response to teachers’—to our—entirely irrational demands.

Other coping mechanisms include reading a translation and trying to match the English with the Latin, which considerably lessens the pain of looking up every other word. Or using Perseus to look up and parse words more quickly, sometimes checking in with the English translation to boot. Again, these are reasonable and rational expedients. What would be irrational is to start at the beginning of the sentence and look up each unknown word as it comes. In each case you’re confronted with a variety of possible meanings: pick the wrong one, and you’re off down a garden path where every subsequent choice will probably be wrong too. And so you have to juggle many possible meanings in your head for each word, and with each subsequent word the decision tree grows more branches. Maybe you can manage for a sentence. But a paragraph? A page? You’re almost certain to go off the right track. And

³¹ For a specimen of deprecation, also demonstrating the method’s prevalence, see Hoyos 1997: 9–10.

then all your effort is wasted. This supposed ideal “reader” is simply bashing their head against a wall.³²

We can debate what the value of Classics is, and why we should or shouldn’t read Latin literature. But whatever the value of Classics may be, it almost certainly does *not* consist in looking up words in the dictionary for hours and days and weeks of your life. But that, in combination with the grammar grind, is what we tend to propose to students. An analogy may be helpful. Let’s say you can bench press 135 pounds, and you want to bench press 225. You don’t start with 225 pounds and get stapled to the bench over and over again. If you tried to lift weights like this, you’d never get to your goal. You’d just get discouraged and probably quit, or maybe you’d find ways to “cheat.” But if you actually wanted to get stronger and lift the weights on your own, you’d start off much easier, with weights that you can handle, and you’d progressively build up, increasing the resistance as your strength increases, always trying to push just a little bit harder. And you’d get better, and eventually you probably could reach your goal.

Maybe the case of Latin isn’t quite as bad as that. Because working through a text and looking up words does lead to some knowledge, and rereading and rereading such a text can transform a formerly incomprehensible passage into some comprehended input.³³ And if you do it enough, you *can* eventually get enough input to make headway. To this we would make two points. The first, to our professional colleagues, is to ask a blunt question: after years and years of study, do *you* read Latin as well as you’d like to? But second, and perhaps more importantly, this method of teaching and learning creates a system of

³² One of the referees of this article feels that this approach to deciphering a text is not only rational but may also inculcate transferable problem-solving skills and meticulous attention to detail. This probably reflects a philosophical difference of opinion about why we learn Latin: we would suggest that problem-solving skills and attention to detail can be more profitably cultivated in other ways. But, more practically, the authors find it hard to imagine that many (any?) students as imagined by this referee actually exist. In our experience, students will either (1) resort to coping mechanisms or (2) fail to understand what they’re reading (i.e., “get it wrong”). But if this hypothetical student did exist, doggedly persevering despite difficulties and dead ends, the time and drudgery it would take them to “get it right” would seem to drain all the joy and pleasure from the reading process, which in fact could no longer be called “reading” in any ordinary understanding of the term. (Again, try to picture a third-semester EFL student working through Shakespeare like this: how long might it take them to “read” *Hamlet*?)

³³ While such rereading is useful, perhaps even necessary, for multiplying Latin input, it doesn’t give the diversity of contexts and the other benefits that you get from repeatedly encountering a word in wider reading and listening.

inequity and differential access. Even if you eventually do get enough input, it takes years and years. So who benefits from such a system? Those who start young. Those who have abundant time (no non-Classics jobs or other responsibilities). Those whose bodies and brains are naturally disposed to this highly unusual way of learning. In sum, those who look a lot like many professional Classicists. And who doesn't benefit? Just about everyone else, i.e., the vast majority of people. Much of our teaching is exclusionary, not because there is anything inherently exclusionary about formal knowledge of grammar, but because "learning to read Latin" on a system like this requires years and years that most people just won't have.³⁴

PART II: CHANGES

It's hard to know 95–98% of the words in a classical Latin text, and the way we tend to teach isn't making it any easier. So what should we do?

Option one is to change nothing. This is not unreasonable if you believe that Latin *cannot* (or should not) be taught as a language. Oscan and Umbrian, for example, can probably only be painstakingly translated, and you might think Latin falls into that category too. In this case the results of SLA research aren't applicable, because you're not teaching a language. But Latin, with its relatively abundant corpus of texts and unbroken tradition of use from antiquity to the present, seems like a language to us.

Option two is simply to tell students to give up. Reading Latin is impossible, so why bother trying? This is not entirely wrong. We do think you should give up on the idea of reading classical Latin as fluently as you can read your native language, and we don't think we should offer students the false promise of reading fluency.³⁵ But that's not a reason to despair, or at least not necessarily a reason to despair. It's just part of the joy and challenge and frustration of learning Latin. We can certainly get meaning out of Latin texts—and these

³⁴ On this type of exclusionary teaching, cf. Keeline 2021: 45–47.

³⁵ Which we do all too often: teachers on the internet peddling (or boasting of) "fluency" when they hawk their wares, university course descriptions with inflated claims ("by the end of the course students should be able to read, with the help of a dictionary, continuous passages from such authors writing in classical Latin as Caesar, Cicero, Virgil, and Ovid"), etc. Even the sober-minded *Standards for Classical Language Learning* imply that such fluency is an achievable goal for advanced students (American Classical League 2017: 10–12). If we felt bound by truth-in-advertising regulations, we might instead tell students: "If you work very hard for years, you might eventually be able to read some Latin texts with the help of a variety of philological aids, like dictionaries, commentaries, and translations."

are certainly texts that many people have found deeply meaningful over the past two thousand years—and we can certainly get better at it.

Option three, of course, is to start making changes.

We should acknowledge straightaway that there is no one-size-fits-all set of changes to implement always and everywhere. Latin learners come in all ages and backgrounds, as do their teachers, and what works for a middle schooler in Missoula might not work for a middle-aged autodidact in Mumbai. Furthermore, different learners have different goals. People should do whatever works for them, and the changes we suggest below aren't intended to be a universal prescription. That said, many of our suggestions are general in scope, and we hope they are applicable, perhaps with modifications, to a variety of educational contexts.

The first clear and obvious change to make is to emphasize vocabulary acquisition at all levels of the curriculum. If you don't know the words in a sentence, you can't read it. Students should try to develop a working vocabulary of thousands of words, much more than they learn from an introductory Latin textbook; learning new words is the one thing that will have the biggest payoff in reading and understanding Latin texts.³⁶ How many advanced Latin classes today really focus on vocabulary? Even at the intermediate level, particularly in college classrooms, we tend not to emphasize this key element. Focusing on vocabulary has a pleasant ancillary benefit too: if we also discuss English derivatives, we not only cement the Latin words but also increase students' command of English in the process.³⁷

The corollary of this recommendation is more controversial: we should probably de-emphasize grammar. Students do need to know grammar in order to read, but our typical ratio of instructional time seems wrong. We ought to rethink what grammar we teach and why we teach it. If we want students to

³⁶This point is well known in SLA research (e.g., Folse 2004: 19–34), and has been made (and largely ignored) by Classicists too: see, e.g., Muccigrosso 2004: 409 (vocabulary is “the single best predictor of reading success”); Carlon 2013: 109 (“vocabulary is critical and dominant”); Gruber-Miller and Mulligan 2022.

³⁷Knowledge of a language influenced by Latin—whether English or a romance language—can be very helpful in acquiring Latin vocabulary and in guessing the meaning of otherwise unknown Latin words (and vice versa!). On the other hand, students whose native language is Chinese, say, face a correspondingly higher barrier to learning Latin vocabulary and trying to read Latin texts. To our knowledge no one has tried to measure these differential challenges for Latin, but the helpfulness of cognates in vocabulary learning more generally is well known: see, e.g., de Groot and Keijzer 2000. Knowledge of etymology and historical linguistics also seems to contribute to vocabulary knowledge and retention: see, e.g., Stratton 2022 (a study of English-speakers learning German).

develop sophisticated and accurate mental representations of Latin, explicit grammatical instruction is probably useful mostly to promote “noticing” of grammatical phenomena in texts that students are reading.³⁸ Otherwise such instruction may have its benefits—many of us philologists find it intrinsically interesting and worthwhile for other reasons, even fun—but it probably won’t increase reading abilities.

Other changes are small and easily implemented tweaks. For example, we ought to use the results of SLA research on vocabulary acquisition when we can.³⁹ So we probably shouldn’t try to learn the names of “all the vegetables” as a semantic set, but rather learn words in meaningful contexts (“a *rabbit hops* through a *garden* and eats *carrots* and *lettuce*” as part of a longer story about Peter Rabbit evading the wrathful rake of Mr. McGregor).⁴⁰ We probably shouldn’t try to memorize bare lists of vocabulary.⁴¹ We probably shouldn’t try to produce words before they’ve been internalized; input precedes output. (This has important ramifications for how we teach prose composition and

³⁸ We hope to discuss the role of grammar in Latin instruction in greater detail elsewhere. Some grammar is essential to reading Latin and would be otherwise foreign to English speakers: agreement of nouns and adjectives, for instance, or recognition of subjects and direct objects. Other grammatical knowledge has relatively little payoff: e.g., the classification of conditional sentences. We should emphasize the fundamental grammatical knowledge that yields massive returns, and defer detailed discussion of complexities and oddities and exceptions. Such details rarely derail a student’s reading experience; students struggle more often with the basics and, especially, with vocabulary. For some sensible remarks on finding *aurea mediocritas* in teaching Latin grammar, see Carlon 2013: 109–11; for an attempt at putting these precepts into practice, see Libatique and Machado 2021, suggesting that we introduce grammatical topics based on “frequency, complexity, and familiarity” (37; see also 46–47 on focused vocabulary acquisition).

³⁹ It’s easy to drown in the deluge of recent research on this topic, but fortunately many scholars take the time to synthesize and summarize their results, especially for the use of teachers: see, e.g., Folse 2004; Schmitt 2008; and Barcroft 2016 (from which the advice in this paragraph is drawn).

⁴⁰ The thematically organized Latin word lists of Meissner 1894, for example, are probably not very useful for vocabulary learning and retention; perhaps the approach of Comenius 1658, featuring thematically grouped vocabulary arranged into simple (and illustrated) stories, is somewhat better, although few will be riveted by the contents of the stories themselves.

⁴¹ Though bare lists may be useful for initial contact, to be followed by encounters in meaningful contexts for long-term retention. Such an approach reduces the problem of incorrect guessing; further Folse 2004: 35–45, 71–84.

how it does or does not help students learn vocabulary.)⁴² In sum, there is a small army of smart people who are thinking hard about how best to learn foreign language vocabulary, and we should probably listen to them.

Some tweaks wouldn't be small, but wouldn't really be so hard to put into practice. We could, for example, create a full-featured digital environment to help students learn words in meaningful contexts. Given a set of target words, we could automatically produce comparable passages containing those words from elsewhere in our corpus; for the student studying Vergil's vocabulary, say, we could try to produce parallel passages from Vergil. These target words could then be paired with freely accessible Latin-Latin and Latin-English dictionary definitions. And finally, all this could be combined with vocabulary-learning techniques like spaced repetition based on quiz results: if you miss a word, you probably need to see it more frequently; if you get a word right, you probably don't need to see it as often. Such a vocabulary-learning environment does not yet exist, but it could be created without undue difficulty by joining together existing digital tools.

The changes suggested so far are easy, more shifts of emphasis than anything else. What follows is harder. First, we ought to increase the amount of time spent at the intermediate level. Right now we tend to have a fanatical commitment to getting students to read "authentic" Latin as early as possible, and for understandable reasons—that's the uniquely valuable thing we have to offer. You could learn Spanish without ever intending to read Spanish literature, but without texts in Latin, there's not much reason to learn the language.⁴³ And

⁴² "Prose composition" can mean many things (translation or original composition? sentences or paragraphs or pages? imitations of a particular author or a generic "classical Latin" style? a focus on particular grammatical topics or expression of meaning?), and a full discussion lies outside the scope of this paper. But observe that, from a vocabulary-learning perspective, writing sentences or extended compositions is likely to be helpful in order to try to cement words *that the student has already been exposed to and is actively trying to learn*. On the other hand, students are unlikely to learn new vocabulary by looking up English words and mechanically substituting Latin equivalents, especially when they have to look up almost every word and contend with grammatical constructions at the limits of their capabilities. (Thus students using *Bradley's Arnold* tend to spend an entire semester mired in the glossary in the back of the book.) Much current SLA research on the task of writing sentences featuring target vocabulary items falls under the broader heading of the "Involvement Load Hypothesis," on which see the recent meta-analysis of Liu and Reynolds 2022 with references and discussion.

⁴³ Some students may profess to have other reasons to want to learn Latin—the devout Catholic who wants to know the language of the Church, the historical linguist interested in reconstructing Proto-Indo-European, the singer who wants to know more about what they're singing—but even these aims depend at some remove on texts written in Latin. If

yet after a year of college study, for example, essentially no human being can read classical Latin, and it is at best inefficient to try to force them to do so. Instead, at this level we should incorporate many more “non-authentic” texts for extensive reading: readers, novellas, and eventually medieval and early modern Latin.⁴⁴ Ideally we want students to be reading hundreds of pages a week. This doesn’t mean that students should *only* read such texts; it means rather that they should primarily focus on those texts, and instructors should use them as introductions to smaller, carefully curated passages of classical Latin. In this way students still get access to the supposed Good Stuff.⁴⁵ As students progress, they can eventually read more and more classical Latin, just as EFL students eventually can read and enjoy Shakespeare, but both groups must pass through a long apprenticeship at the intermediate level first. The adjustment of the ratio of extensive and intensive reading, especially at the intermediate level, is probably even more important than the adjustment in emphasis accorded to vocabulary and grammar.

We should also try to incorporate other sources of comprehensible input, especially outside class. YouTube videos or podcasts in Latin; Latin songs; Latin games; anything. None of these require that a teacher learn to speak or write Latin; the materials are already out there.⁴⁶ In class, on the other hand, a major imperative should be to get more input. Translating a single paragraph of Latin into English results in very little actual input. Even if such translation remains a major component, or even the core, of a class session, a few additional activities can add a *lot* of repetitions of vocabulary in con-

all English texts disappeared today, English would still exist. If all Latin texts disappeared today, Latin would disappear with them; even if a few people continued speaking it, without reference to a fixed canon of *Latinitas* the language would soon begin to change and wouldn’t be Latin anymore. None of this is to deny that there are side benefits to learning Latin, like better understanding of other languages or of language in general, but these side benefits flow from our corpus of surviving Latin texts.

⁴⁴ The only way anyone could *ever* hope to be exposed to tens of millions of words of Latin—the theoretical road to fluency (cf. n27 above)—is to read postclassical Latin, where in fact billions of words are readily available, many of which are fascinating and deserving of being read in their own right, even if they weren’t written by native speakers.

⁴⁵ From a vocabulary perspective, students should begin with classical Latin prose texts and proceed only later to poetry. Unfortunately but inevitably, the Latin texts that are most compelling tend to have the greatest variety of vocabulary, and the texts that are the simplest to read tend to be, at least in the eyes of many students, less compelling. But if you take primarily vocabulary into account, you might suggest a progression beginning with *Nepos* and *Caesar*.

⁴⁶ For Latin podcasts and YouTube channels, see the collections at <https://scholaeinterretiales.wordpress.com/teach-yourself-latin/>.

text. Consider something as basic as “Simon says”: in its simplest version, you might have something like “Simon says touch your *nasus* [nose].” Later you can return to this activity with “*Simon dicit: tange nasum*” (“Simon says: touch your nose”). Finally you can say things like “*Caesar te iubet nasum tangere*” (“Caesar orders you to touch your nose”). The point, in any case, is to get more comprehensible input in meaningful contexts. So too with various contemporary “comprehensible input” activities like circling or Movie Talk.⁴⁷

Another relatively small change with relatively big benefits is to increase the amount of time spent doing pre-reading activities.⁴⁸ Introductory summaries of a passage, in English or Latin, are extremely helpful. Scaffolded readings, in which students read simplified versions of the Latin text that get progressively more complicated as they approach the original, are also very helpful, and such things already exist for many texts.⁴⁹ Still, of course, when students come to read the original Latin text, they won’t know every single word. And the closer at hand help is, the less interrupted the reading experience: marginal glosses are ideal, especially if they’re in Latin, which further increases input.⁵⁰ At all events

⁴⁷ Such activities are most easily explored on the internet; see, e.g., <https://comprehensibleclassroom.com/how-to/essential-strategies-for-tpsci-teachers/how-to-circle/> and <https://comprehensibleclassroom.com/about-the-comprehensible-classroom/how-to/movietalk/> for circling and Movie Talk, respectively.

⁴⁸ The case for the importance of pre-reading for Latin has long been made (cf., e.g., Rea 2006); again, today such activities are most easily explored online (see, e.g., Toda n.d.).

⁴⁹ Helpful discussion of how to create and use “tiered texts” in the Latin classroom can be found in Sears and Ballestrini 2019; although the authors imply that this is a new strategy borrowed from modern language instruction (72), in fact it was already a standard part of Renaissance Latin classrooms. Some scaffolded texts can be found among the resources collected at <https://vivariumnovum.it/scolastica/subsidia/docenti>, and note that the verse texts in the *Bibliotheca classica* and Delphin editions series (see next note) contain prose paraphrases that amount to tiered readings as well.

⁵⁰ There is a large body of literature on the efficacy of different forms of glossing, and it must be admitted that putative best practices are not yet clear (recent review in Yanagisawa, Webb, and Uchiyara 2020). If the gloss itself isn’t understandable, it’s useless; for that reason there may still be a place for glosses in a learner’s native language. A massive list of online editions of texts with Latin–Latin glosses in the *Bibliotheca classica Latina* and *Scriptores Latini in usum Delphini* series can be found at <https://subsidia.vivariumnovum.it/risorse-didattiche/per-la-pratica-didattica/classici-latini-in-edizione-monolingue>. Latin marginal annotations (often illustrated) are the hallmark of Hans Ørberg’s *Lingua Latina per se illustrata* and its supplements (<https://www.hackettpublishing.com/lingua-latina-per-se-illustrata-series/readers-and-supplements>); Geoffrey Steadman has a number of free commentaries with same-page English glosses and commentary, and he has at least once experimented with Latin–Latin glosses as well: see Steadman 2019. Programs like The Bridge (Mulligan et al. 2014–) make it much easier for teachers and commentary writers to know what they might want to gloss.

we should eschew the policy exemplified by E. J. Kenney's pronouncement in designing the Cambridge Greek and Latin Classics series: no vocabularies were to be included, because "the students ought to use dictionaries."⁵¹ For all but the most advanced students, the dictionary should be the resource of last resort: it is inefficient, disruptive of the reading experience, and frankly hard to use. (Or, if easy to use, then pointless: why should students have to consult a separate weighty tome to find a one-word gloss?)

Larger changes could produce larger benefits. Converting to entirely communicative classrooms is the obvious example, with the potential to expose students to much larger amounts of comprehensible input, but it's equally obvious that this is a major challenge for most teachers. Even teachers who prefer to stick strictly to translating from Latin into English could still make changes that would result in more efficient uses of time. Consider the tedium of a typical translation session: students fumble through a text that they have supposedly read the night before; teachers spend much of their time correcting errant translations and asking for grammatical information; non-translating students stare out the window; very little is "read," and what is read is read in plodding fashion. Much of this difficulty arises because students haven't really understood what they "prepared" at home; if they had, there wouldn't be much point to working so hard to translate it again in class.

But the translation classroom can easily be "flipped." Students could read *new* texts in class with their teacher, who can then serve as an interactive dictionary and grammar and commentary all rolled into one.⁵² The teacher thus provides students with continuous and immediate feedback on their understanding of the text, tailored to their precise difficulties.⁵³ These students then spend their time at home *rereading* the text that they translated in class that day; what would have been largely incomprehensible to them if they were slogging through it for the first time on their own has suddenly become comprehended input.⁵⁴ And rather than spending two hours preparing the text in

⁵¹ Quoted in Gibson 2021: 181.

⁵² This is hardly a modern innovation; indeed, before widely available printed books, teachers were inevitably forced to expound on texts that their students had not read in advance. Calls for a similar method have been made by others (e.g., Distler 1962: 89–105), but the idea does not seem to have been widely adopted. For the somewhat different task of "flipping" beginning Latin classrooms, see Harvey 2014; Natoli 2014.

⁵³ The importance of immediate and tailored feedback for successful learning is well known: see, e.g., Ericsson and Harwell 2019: 3–4. The delay in feedback when a prepared translation is corrected a day or days later is thus a significant obstacle to efficient language acquisition.

⁵⁴ Skilled learners reread as a matter of course; others will need motivation. A simple motivator, and a useful companion activity, is the completion of a reading comprehension worksheet; for an example, see Keeline 2019b: 61–62.

vain and another hour having their translation corrected, they can cover very nearly as much text in class and have lots of time left over for other things. This minimizes struggle and frustration, and leaves students time at home to work on pre-reading activities and extensive reading and other things *at their level*.⁵⁵ One of the authors of this paper has adopted this method in his undergraduate teaching, and while there is no royal road to learning Latin, he has gotten positive feedback from his students and enjoyed the process himself. Another of his colleagues has used a mix of prepared and unprepared reading with good success, and such a *via media* may prove more attractive still.

Again, every teacher has to find approaches that work for them and for their students. Whether any of these specific suggestions constitutes such an approach, there seems little reason beyond tradition to continue as we have been doing things.

CONCLUSION

“There seems little reason beyond tradition to continue as we have been doing things”: this holds true for much of Latin instruction. We can do better, and in many cases we can do better without even working harder or having less fun. Less work, more fun, better understanding of Latin: where’s the catch? There is no catch, except this: vocabulary is only one of many challenges that readers face when confronted with a Latin text. It’s *the* fundamental challenge, but many other obstacles remain, and, as we’ve seen, the vocabulary obstacle is ultimately insuperable. So none of this is going to magically make students able to read Latin with ease and pleasure.⁵⁶ Maybe no one can really do that. But we *can* get closer, and we *can* choose to be more or less efficient in our approach to teaching and learning. And, whether you want to expand access to Latin or fight declining enrollments or just optimize your instructional time and your students’ learning experience, we have every reason to try to do better.

⁵⁵ Some teachers may object that their goal is to make students *independent* readers—even if they can’t be fluent—and that the procedure suggested here would make students entirely dependent on their teachers. But the traditional approach, in fact, makes students permanently dependent on lexical aids and other crutches (almost certainly including translations). Because such students cannot get enough input, they cannot hope to acquire the language, and in practice they themselves remain forever dependent on their teachers’ corrections (again, if they were not dependent on those corrections, there would be no point to reviewing translations in class). Students can only become independent of their teachers if they actually acquire the language, and language acquisition happens through massive amounts of comprehensible input: there are no shortcuts. While it requires an initial investment of time, this approach is in the end a more efficient way to promote independent reading.

⁵⁶ Cf. Keeline 2019a, 2019b: 60–61.

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